



RossSuper 500

High performance descaling acid

RossSuper 500 is a nitric acid based product recommended for descaling equipment, tanks, lines and pipes by the circulation method (CIP) in Beverages, Dairy, Refrigerators and Food Industries in general. It is also used in acid rinsing and as passivation agent for stainless steel installations.

RossSuper 500 is an acidic product, predominantly of Nitric Acid, with steam. The continuous use of product, after the alkaline step of the CIP process, inhibits the formation of inorganic deposits, and can be used at high temperatures for their removal. Product is highly effective in the removal of most inorganic deposits, especially Calcium or Magnesium Carbonate and Phosphate. Under the conditions of use it does not form foam in any circulation process.

Key Benefits

- Highly aware of inorganic soil removal
- The foam breaks down very quickly preventing buildup in tanks, lines and pumps
- Dosage and concentration control can be conductivity, even at low concentrations
- Removal and hunting of milk stones and inorganic inlays in general
- Total surface humidification
- Can be dosed automatically by automated systems

Application Method

For CIP sanitation in Beverages Industries use concentration between 2.0 and 3.5% (w / v), after rinsing the alkaline solution. Circulate for 30 to 50 minutes at room temperature.

For acid rinsing use concentration between 0.4 and 0.8% (w / v). Circulate for 15 to 30 minutes. For passivation use concentration of 30 and 40% (w / v). Circulate for 2 hours at a temperature between 50 and 60 ° C.

For CIP descaling in Food Industries in general, circulate the solution of the product in the concentration between 1.0 and 1.5% (w / v) for 20 minutes at 60-65 ° C.

Important: Acid detergents should not be circulated at temperatures above 70 ° C where release of gases and vapors which substantially increase the possibility of corrosion, even on stainless steel surfaces.

Product solution can be reused, however before any new its concentration should be analyzed and strengthened, if necessary. The reuse depends on the facilities and operating conditions of each process. Always rinse the equipment with potable water. The concentration of the ideal concentration, temperature and contact time will depend on the application.

For more details, please contact your local Buzil Rossari representative.



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Technical Data

Appearance	Clear colorless to yellowish liquid
pH	$< 1.5 \pm 0.3$
Specific weight at 25°C	1.31 gm/ml
Total Acidity (% HNO ₃)	51.5

Test Method

Reagents & Indicators: 0.1 N Sodium Hydroxide solution, Phenolphthalein (1% in neutral ethyl alcohol)

Method: Add 2 to 3 drops of indicator Phenolphthalein. Titrate with 1.0 N NaOH until colorless to pinkish.

Calculation:

%w/w RossSuper 500 = volume (mL) x 0.13

%v/v RossSuper 500 = volume (mL) x 0.10

Compatibility

Product can be used on most materials encountered in the food industry, if the product is applied according to concentration and temperature recommendations. It is recommended to carry out tests for other materials before prolonged use.

Storage & Handling

All safety information on the handling and use of this product is provided in the Safety Data Sheet which can be consulted and / or obtained from our representatives

Keep this product in the original packaging or, where possible, in a tank of bulk reception, protected from extreme temperatures.

Declaration: Technical figures given in this data sheet are typical values based on internal laboratory measurements, following appropriate International Standard Methods, and should not be taken as a warranty or representation. It is the end user's responsibility to make sure that the product matches its requirements.

For more information on this product, please contact your local Buzil Rossari representative.

Buzil Rossari Private Limited

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Food Grade Quality Certificate

Application in food & beverage processing facilities

Here we confirm that, RossSuper 500, if used according to the instructions in the product information sheet is suitable for the open plant cleaning in Beverage, Brewing, Dairy & Food Processing facilities and not to adulterate the product in any manner during the use.

Essential condition is a thorough rinsing with clear portable water of surfaces intended to get in direct contact with foodstuff.

The remaining residuals that may contaminate or adulterate food contacting cleaned surfaces are expected to be negligible, following a potable water rinse. Therefore, the potential ingestion of residues resulting from the approved use of these types of products does not represent a significant health hazard after our assessment of the products as suitable for use under the intended conditions.



Thanking you,

Yours faithfully,

For Rossari Biotech Ltd.

Please note that this is not a regulatory assessment and should not be construed as an endorsement of regulatory compliance.



User Concentration Test Procedure/Method

Product	RossSuper 500	Issued On	01/01/2018
Application	Heavy duty non-fuming acidic descaler	Next Review On	01/01/2023

Reagents

DM Water, 0.1N NaOH Solution, Phenolphthalein Indicator

Apparatus & Instruments

Conical Flask, Measuring Cylinder

Method

Take 50mL of RossSuper 500 sample in a conical flask

Add few drops of Phenolphthalein Indicator

Titrate against 0.1N NaOH till stable pink

Take a note of burette reading.

Calculation

% Activity : Burette Readings in mL X 0.24

Where, BR is burette reading in mLs of 0.1N NaOH.

0.24 is the calculation factor.

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

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1. Identification of the mixture and of the company / undertaking

1.1. Product identifier

RossSuper 500

1.2. Relevant identified uses of the mixture and uses advised against

Professional cleaning/maintenance product for food & beverage industries.

1.3. Details of the supplier of the safety data sheet

Manufacturer:

Manufactured under licence from buzil
ROSSARI BIOTECH LIMITED
Plot No. 10 & 11, Survey No.90/1/10 & 90/1/11/1,
Kumbharwadi, Village Naroli
Silvassa-396235
Dadra and Nagar Haveli (UT)
India

e-mail info@rossarimail.com

www.buzil-rossari.com & www.rossari.com

1.4. Emergency telephone number

+91-22-61233800/61233887

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2. Hazards identification

Risk phrases :

Hazards for man and the environment : R34 - Causes burns.

Classification system : The classification complies with current EC regulations on dangerous substances and preparations.



3. Composition / information on ingredients

Chemical characterization

Description : Mixture in water of non-hazardous ingredients and the substances listed below.

Hazardous ingredients :

MAJOR/HAZARDOUS INGREDIENTS NAME	CAS NUMBER	APPROX. PERCENTAGE
Nitric Acid	7697-37-2	≥ 30%

Additional information : The full text of the relevant risk phrases is given in section 16.

4. First aid measures

General information : Remove contaminated clothing immediately.

Inhalation : Remove from source of exposure. Obtain medical attention if symptoms develop. In case of unconsciousness bring patient in to stable position for transport.

Skin contact : Wash skin immediately and thoroughly with water. Remove contaminated clothing. Obtain medical attention if symptoms develop.

Eye contact : Wash immediately with copious amounts of water and obtain medical attention.

Ingestion : Remove material from mouth. Drink one or two glasses of water (or milk) and obtain immediate medical attention.

5. Firefighting measures

Suitable extinguishing media CO₂, extinguishing powder or water jet. Fight larger fires with water jet or alcohol resistant foam.

Protective equipment : No special measures required

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6. Accidental release measures

Protection of personnel : Wear suitable protective clothing, gloves and eye / face protection.

Environmental precautions : Dilute with plenty of water

Method of disposal : Absorb with liquid binding material (sand, diatomite, universal binders, sawdust).

Use neutralizing agent, Absorb onto dry or similar inert material. Dispose of the collected material according to the regulations.

7. Handling and storage

Handling (see also sections 8 and 15)

Information for safe handling : Use common rules for working with chemicals.

Information about protection against explosions and fires : No special measures required.

Storage :

Requirements for storage rooms and containers : According to local legislation.

Storage in one common storage facility : According to local legislation.

Further information about storage conditions : None.

8. Exposure controls / personal protection

Additional information about design of technical systems : No further data; see section 7.

Components with limit values that require monitoring at the workplace:

CAS: 7697-37-2 Nitric acid

OES (Great Britain) Short-term value: 10 mg/m³ , 4ppm

Long-term value: 5.2 mg/m³ , 2ppm

Additional information : The lists that were valid during the compilation were used as basis.

Personal Protection equipment :

General protective and hygienic measures :

Keep away from foodstuffs, beverages and food. Take off immediately all contaminated clothing.

Wash hands during breaks and at the end of work. Avoid contact with the eyes and skin.

Breathing equipment : Personal protection is not normally required.

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Protection of hands : Chemical-resistant protective gloves made of butyl or nitrile rubber (EN 374, category III). In consultation with the supplier of protective gloves a different type of similar quality may be chosen.

Eye protection: Tightly fitting safety glasses

9. Physical and chemical properties

Appearance: Clear uncoloured to yellowish liquid

Odour: Characteristic

pH : 1.5 ± 0.3

Melting point / melting range: ND

Boiling point / boiling range: ND

Specific gravity (25 °C): 1.131

Solubility in / miscibility with water: Miscible in water

Viscosity (25 °C): ND

10. Stability and reactivity

Thermal decomposition / conditions to be avoided : No decomposition if used according to specifications.

Dangerous reactions : Reacts with alkali and metals

Hazardous decomposition products : No dangerous decomposition products known.

11. Toxicological information

Acute toxicity:

LD50 (oral) : Based on the toxicological classification the LD50 (oral) is estimated to be 200-2000 mg/kg. This value is of no practical significance due to the corrosive nature of the product.

Primary irritant effect:

Skin contact: Causes burns

Eye contact: Causes severe or permanent damage

Inhalation: Severe irritant, may cause pulmonary oedema.

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Ingestion: Causes severe burns. Ingestion will lead to strong caustic effect on mouth and throat and throat and to the danger of perforation of esophagus and stomach.

Additional toxicological information:

EC classification: Corrosive.

12. Ecological information

General notes : Must not reach sewage water or drainage ditch undiluted or unneutralized. When used for its intended purpose, this product should not cause adverse effects in the environment. The information in this section refers to the undiluted product.

13. Disposal considerations

Product Recommendation : Dispose of observing official regulations.

European waste catalogue : 20 01 29 detergents containing dangerous substances

Uncleaned Packagings Recommendation : Dispose of observing official regulations.

Recommended cleaning agent : Water, if necessary with cleaning agent..

14. Transport information

Land transport ADR/RID (cross-border)

ADR/RID Class : 8 Corrosive substances

Kemler Number : 80

UN-Number : 2031

Packaging Group : II

Label : 8

Proper Shipping Name : 2031, NITRIC ACID SOLUTION

Maritime transport IMDG :

IMDG Class : 8

UN-Number : 2031

Label : 8

Packaging Group : II

EMS Number : 8-03

Maritime Pollutant : No

Proper Shipping Name : 2031, NITRIC ACID SOLUTION

Air transport ICAO-TI and IATA-DGR :

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ICAO / IATA Class: 8

UN-Number : 2031

Label : 8

Packaging Group : II

Proper Shipping Name : 2031, NITRIC ACID SOLUTION

Transport / Additional Information : Transport regulations include special provisions for certain classes of dangerous goods packed in limited quantities.

15. Regulatory information

Classification according to EC guidelines : The product has been classified and labeled in accordance with current EC Directives.

Symbol and hazard designation of product:



16. Other information

The information in this document is based on our best present knowledge. However, it does not constitute a guarantee or any specific product features and does not establish a legally binding contract.